Musanna Galib

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Summary

Ph.D. mechanical engineer and materials researcher specializing in energy-storage materials, multiscale computational modeling, scientific data infrastructure, and machine learning for materials. Argonne postdoctoral appointee building FAIR, AI/ML-ready data workflows for ESRA, with experience spanning DFT/MD, finite-element modeling, machine-learning interatomic potentials, HPC, battery testing, and open-source scientific software.

Education

- **The University of British Columbia (UBC)** — Vancouver, Canada
  Ph.D., Mechanical Engineering; 4-Year Doctoral Fellow — *Sep 2020 – Dec 2025*
- **Bangladesh University of Engineering and Technology (BUET)** — Dhaka, Bangladesh
  M.Sc., Mechanical Engineering; CGPA: 3.92/4.00 — *Apr 2017 – May 2019*
- **Bangladesh University of Engineering and Technology (BUET)** — Dhaka, Bangladesh
  B.Sc., Mechanical Engineering; CGPA: 3.92/4.00; Rank: 4/180 — *Apr 2012 – Feb 2017*

Experience

- **Postdoctoral Appointee - Scientific Data Management** — Lemont, IL, USA
  Argonne National Laboratory, Energy Storage Research Alliance (ESRA) — *Jan 2026 – Present*
  - Develop FAIR scientific-data infrastructure for energy-storage R&D, including ETL pipelines, metadata schemas, provenance, and machine-readable experimental records.
  - Standardize heterogeneous battery characterization and simulation outputs into reusable, AI/ML-ready datasets and scientific workflows for the ESRA Data Catalog.
  - Partner with experimental, computational, and software teams on interoperable data services, community standards, and cross-institution energy-materials data integration.
- **Graduate Research Assistant - Materials Modeling & Battery Research** — Vancouver / Kelowna, Canada
  The University of British Columbia — *Oct 2021 – Dec 2025*
  - Led computational-experimental research on metal-anode dendrite suppression and thin-film mechanics, connecting structure–property–performance relationships across atomistic, continuum, and battery experiments.
  - Built multiscale workflows using density functional theory (DFT), molecular dynamics (MD), finite-element modeling, and PyTorch-based machine-learning interatomic potentials; ran simulations on HPC and analyzed data in Python/MATLAB.
  - Performed battery assembly, electrochemical cycling, in-situ optical microscopy, and profilometry; developed open-source tools for atomistic data, dendrite tracking, and MOOSE post-processing.
- **Assistant Professor of Teaching / Lecturer** — Dhaka, Bangladesh
  Bangladesh University of Engineering and Technology (BUET) — *May 2017 – Sep 2021*
  - Combined undergraduate engineering instruction with testing and consulting projects involving automotive components, engines, emissions, fuels, and structural-integrity evaluation.

Selected Technical Projects

- **ESRA FAIR Data Platform** — data.energystoragera.org
  - FAIR scientific data catalog and workflow infrastructure supporting ingestion, metadata harmonization, ontology-aware mapping, provenance, ETL, and reusable services for energy-storage experiments and simulations.
- **AtomProNet - Atomic Data Processing for Neural Networks** — GitHub
  - Open-source Python package for transforming and exchanging atomistic simulation data in machine-learning interatomic-potential workflows; published in *Machine Learning: Science and Technology.*
- **SurfTrack & MOOSEanalyze** — GitHub
  - Open-source tools for automated dendrite tracking from in-situ microscopy and post-processing of MOOSE/Exodus simulation outputs.

Technical Skills

- **Programming & ML:** Python, NumPy, pandas, scikit-learn, PyTorch, MATLAB, Bash, C
- **Computational Materials & HPC:** DFT, MD, AIMD, VASP, LAMMPS, ABAQUS, MOOSE, multiscale modeling, Linux, HPC
- **Scientific Data & Software:** FAIR data, materials informatics, metadata schemas, ontology mapping, provenance, ETL (data pipelines), Git, GitHub
- **Experimental:** Battery cell assembly, electrochemical cycling, in-situ optical microscopy, profilometry, cleanroom workflows

Selected Publications

- **Machine Learning: Science and Technology (2026): Galib, M.**, Isiet, M., and Ponga, M. "AtomProNet: Data flow to and from machine learning interatomic potentials in materials science."
- **ACS Applied Materials & Interfaces (2025): Galib, M.**, Xu, J., Amardeep, A., Liu, J., and Ponga, M. "Dendrite Suppression in Zn Batteries Through Hetero-Epitaxial Residual Stresses Shield."
- **Journal of the Mechanics and Physics of Solids (2024): Galib, M.**, Orhan, K.O., Liu, J., and Ponga, M. "Residual Stress Development in Lattice Mismatched Epitaxial Thin Films via Atomic and Molecular Layer Depositions."
- **Nano Energy (2024):** Xu, J.*, **Galib, M.***, et al. "High-entropy Strategy to Suppress Volumetric Strain and Enhance Diffusion Rate of $Na_3V_2(PO_4)_2F_3$ Cathode for Durable and High-areal-capacity Zinc-ion Battery Pouch Cells." (*Co-first authors)

Full publication list: Google Scholar.

Selected Grants & Funding

- **DRI EDIA Champions, Digital Research Alliance of Canada:** CAD 35,000 program funding.
- **4-Year Doctoral Fellowship, UBC:** CAD 93,267.
- **Collaborative Research Mobility Award, UBC:** 2023.
- **Institute of Applied Mathematics Travel Grant:** 2025.

Honors & Awards

- **1st Prize, Ideas With Impact Competition:** ICICS / UBC, 2024.
- **Best Presentation Award:** CSME Congress in Computational Mechanics, University of Toronto, 2024.
- **President's Academic Excellence Initiative Ph.D. Award:** UBC, 2020–2025.
- **Faculty of Applied Science Graduate Award:** UBC, 2021.
- **Academic Recognition:** Dean's List and University Merit for four consecutive years at BUET.

Leadership & Professional Engagement

- **Vice President (Academic):** Mechanical Engineering Graduate Association, UBC, 2022–2023.
- **Steward:** UBC Teaching Assistants' & Instructors Union (CUPE 2278), 2024–2025.
- **Peer Reviewer:** Additive Manufacturing; Journal of Energy Storage; Journal of Applied Physics.

Additional

- **Professional Training:** NASA Open Science 101; Project Management (Mitacs).